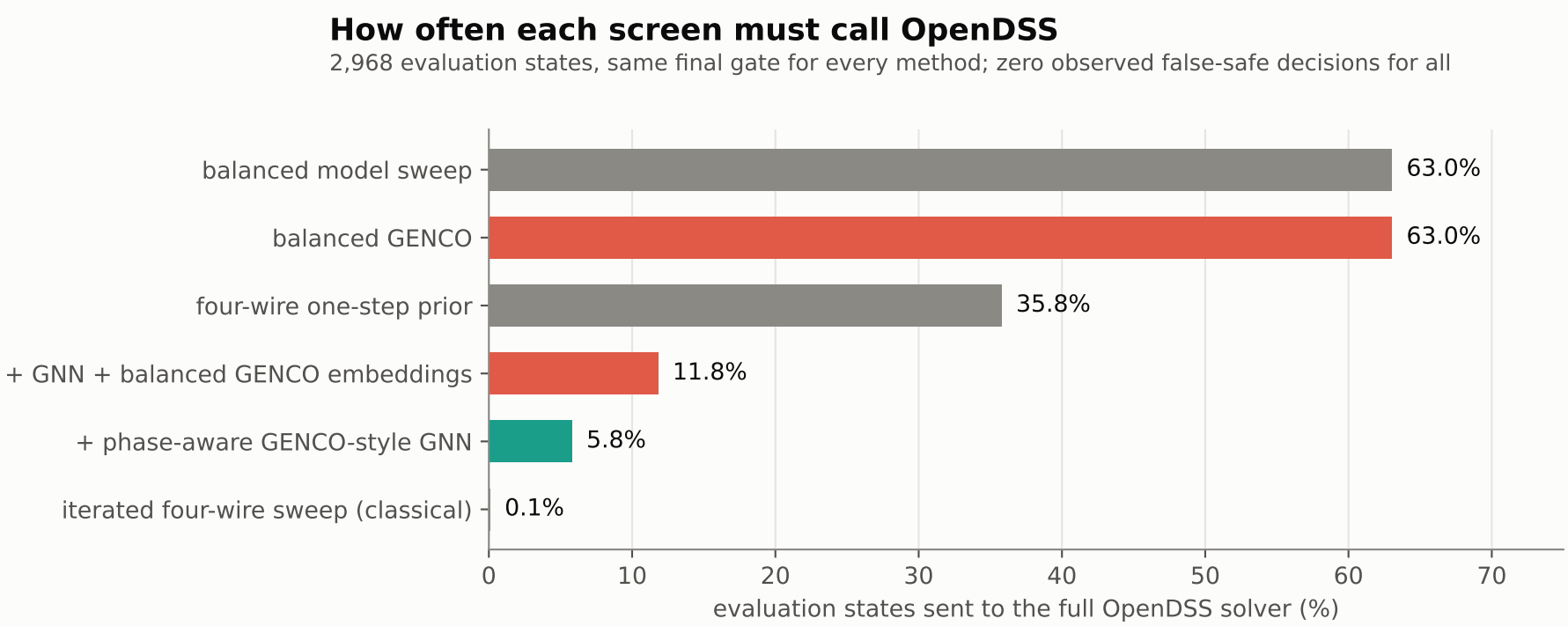
How often each screen must call OpenDSS
2,968 evaluation states, same final gate for every method; zero observed false-safe decisions for all
balanced model sweep
balanced GENCO
four-wire one-step prior
+ GNN + balanced GENCO embeddings
+ phase-aware GENCO-style GNN
iterated four-wire sweep (classical)
63.0%
63.0%
35.8%
11.8%
5.8%
0.1%
0
10
20
30
40
50
60
70
evaluation states sent to the full OpenDSS solver (%)